CYBR 343 W10-Lab

Important Instructions:

- All tools required for this lab are provided in the ZIP file attached with the lab.
- Download the ZIP file and extract all files before starting the lab.
- Do not remove or edit any part of the text. Complete the work in this document and submit a word copy not pdf of this file.
- Only submit this work via Blackboard. I will not accept submissions via email.
- For each part of the lab, include screenshots showing the steps you performed.
- Use the Snipping Tool (or screenshot utility) on your computer to capture screenshots of your work; do not take photos of your screen using a phone or external camera.

Lab Objectives

After completing this lab, you should be able to:

- Hide messages inside files using steganography tools
- Hide files inside images
- Retrieve hidden information from files
- Understand how whitespace steganography works
- Recognize how hidden information may be stored inside normal files

Tools Used

All tools are included in the lab ZIP file.

- SNOW
- S-Tools
- StegoMagic

Part 1 – Steganography Using SNOW

Create a text file named cover.txt with at least 10 lines of text.
Hide the message using the command:

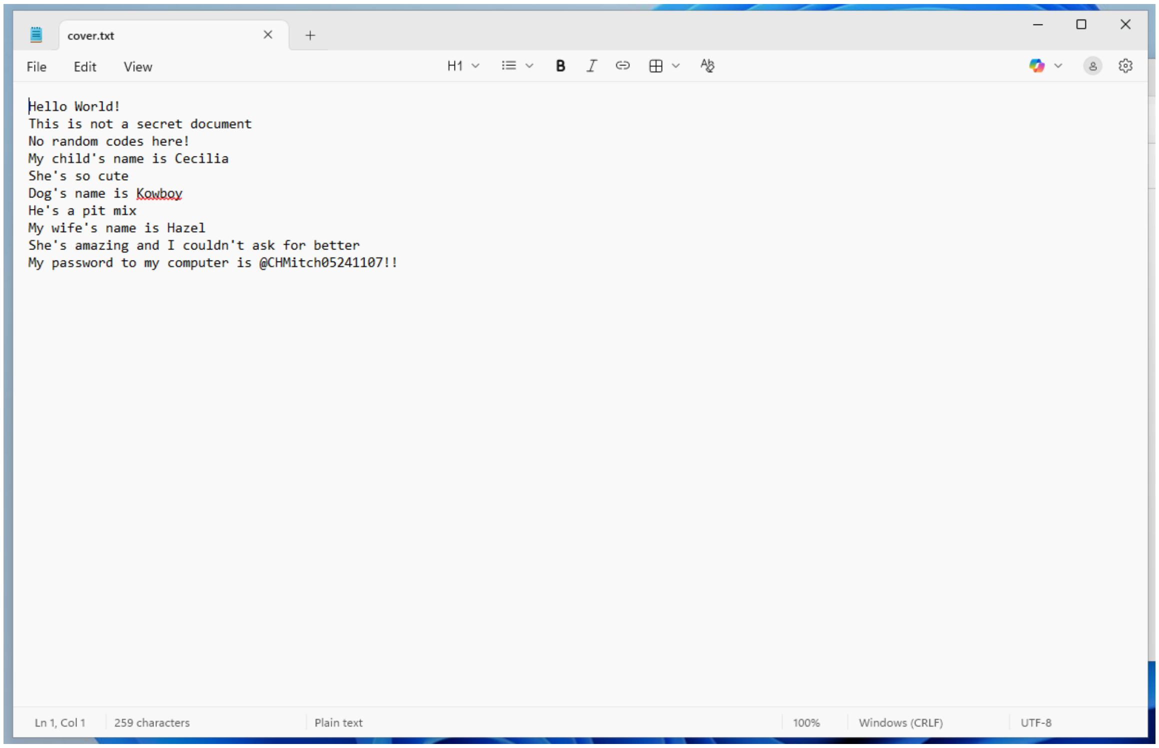
snow -C -p secret123 -m "This is a hidden message" cover.txt stego.txt

Reveal the message using the command:

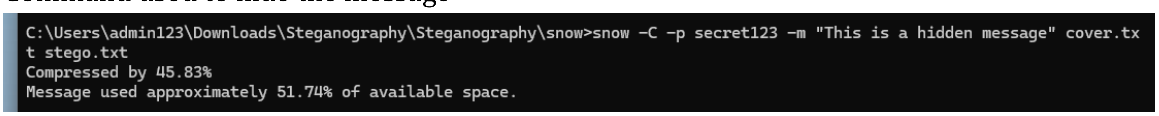
snow -C -p secret123 stego.txt

Screenshots Required :

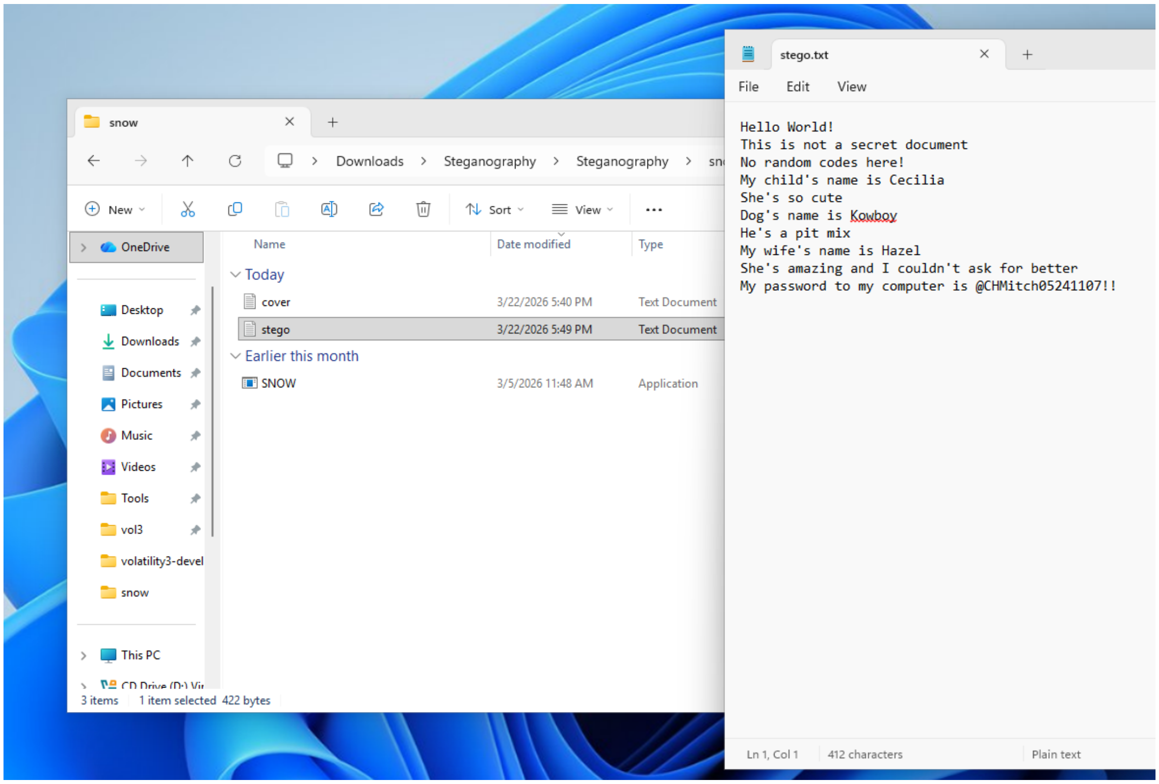
1. Original cover.txt file (in notepad)



2. Command used to hide the message



3. Stego.txt file (in notepad)



4. Command used to reveal the message and revealed hidden message

```
C:\Users\admin123\Downloads\Steganography\Steganography\snow>snow -C -p secret123 stego.txt
This is a hidden message
C:\Users\admin123\Downloads\Steganography\Steganography\snow>
```

5. Output of Dir command in command prompt showing the sizes of cover.txt and Stego.txt. Also discuss what seems suspicious there. Comment on why the size of these two are different.

```
 Directory of C:\Users\admin123\Downloads\Steganography\Steganography\snow

03/22/2026  05:49 PM    <DIR>          .
03/22/2026  05:49 PM    <DIR>          ..
03/22/2026  05:40 PM               269 cover.txt
03/05/2026  12:48 PM            62,464 SNOW.EXE
03/22/2026  05:49 PM               422 stego.txt
               3 File(s)         63,155 bytes
               2 Dir(s)  24,921,182,208 bytes free

C:\Users\admin123\Downloads\Steganography\Steganography\snow>
```

Stego.txt is slightly larger than cover.txt because SNOW hides the secret message using extra whitespace such as spaces and tabs at the ends of lines. They aren't visible when viewing the file in a text editor but still take up file space, increasing the file size.

Part 2 – Steganography Using S-Tools

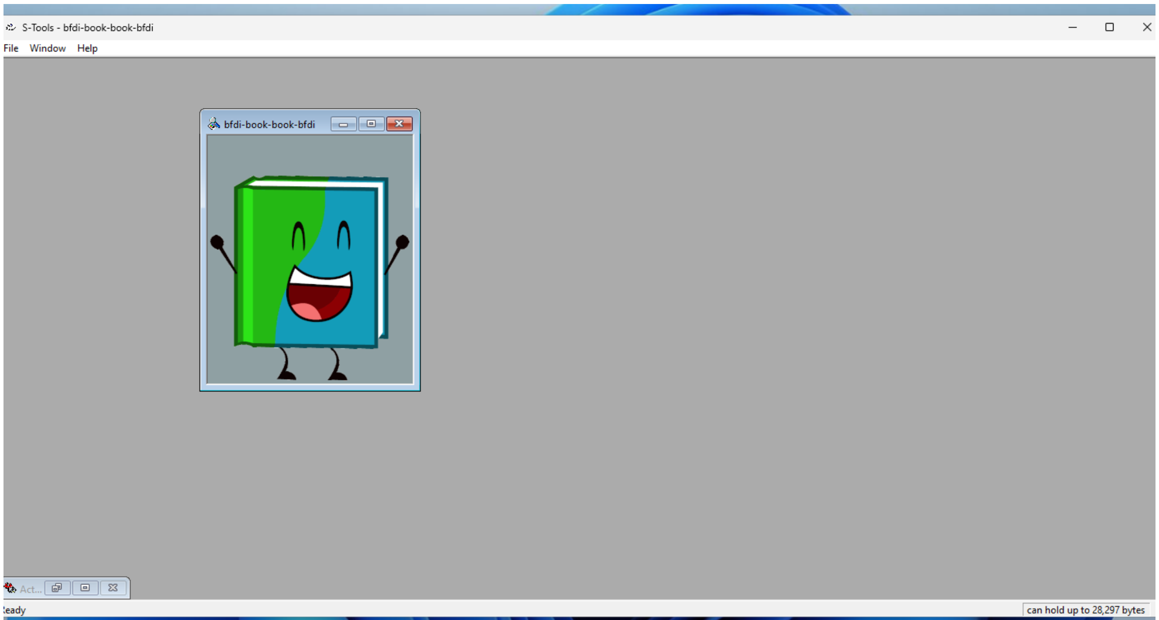
Follow the Lecture 3 Video posted for this week in Blackboard.

Tasks: You are required to hide an image within another image and a text file within an image. You can use images that you like. Make sure images are in .bmp format.

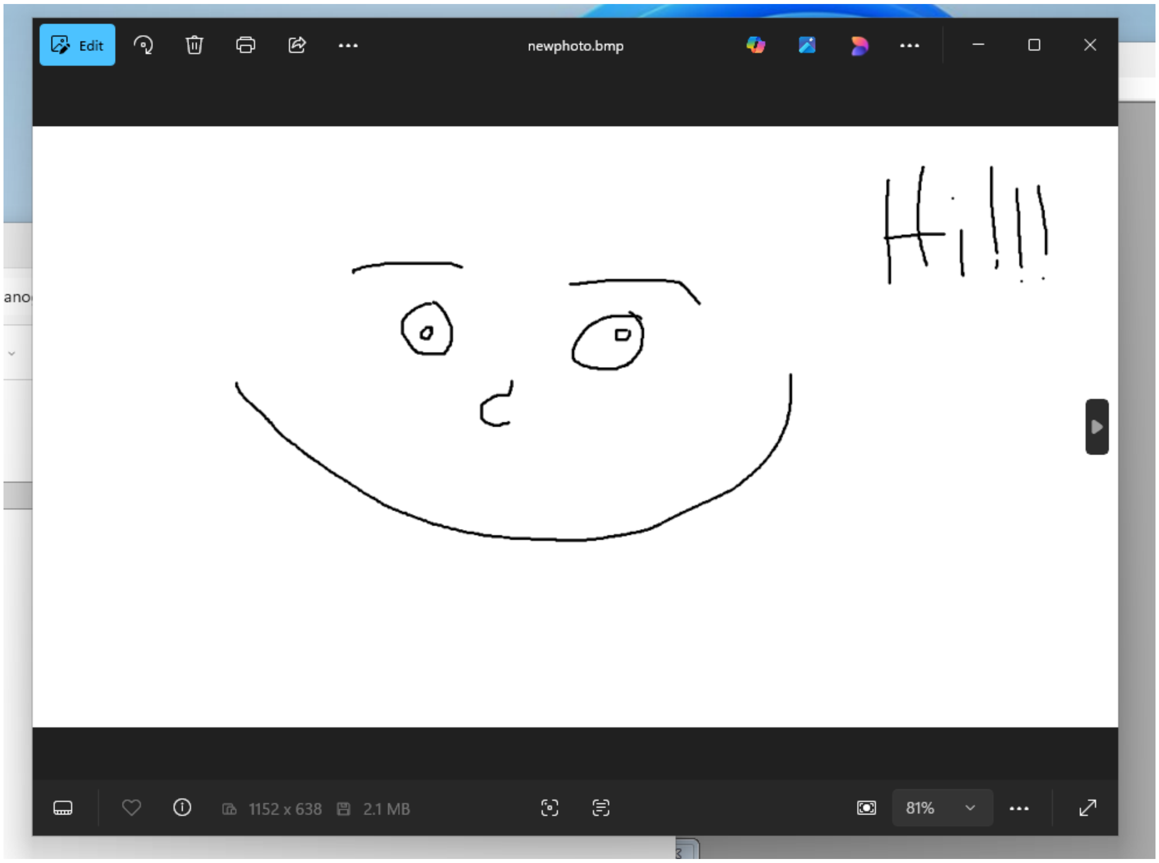
Task 1 – Hide a Picture Inside Another Picture

Screenshots Required:

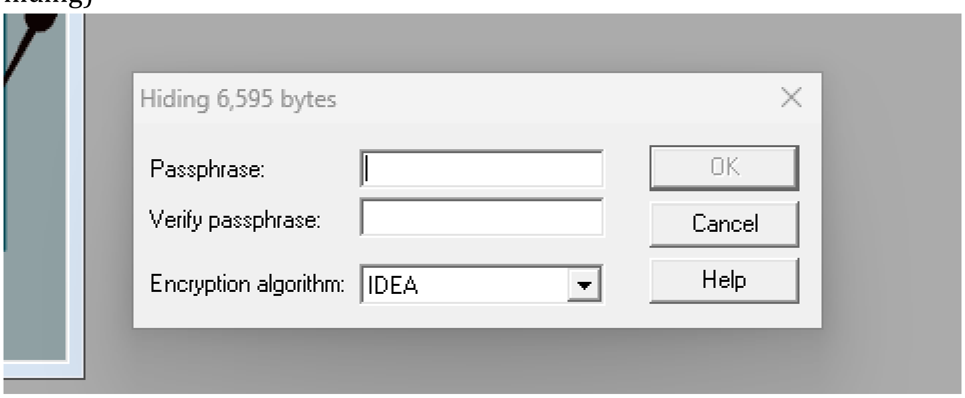
1. Cover image loaded



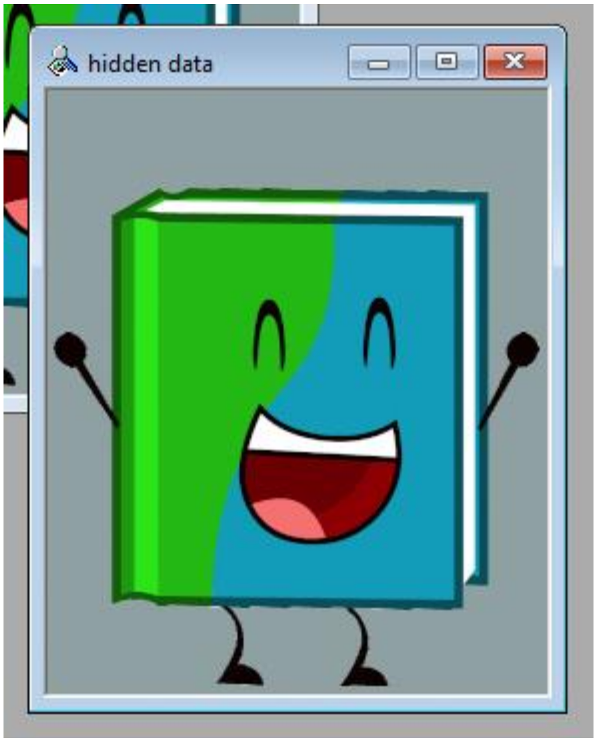
2. Image being hidden



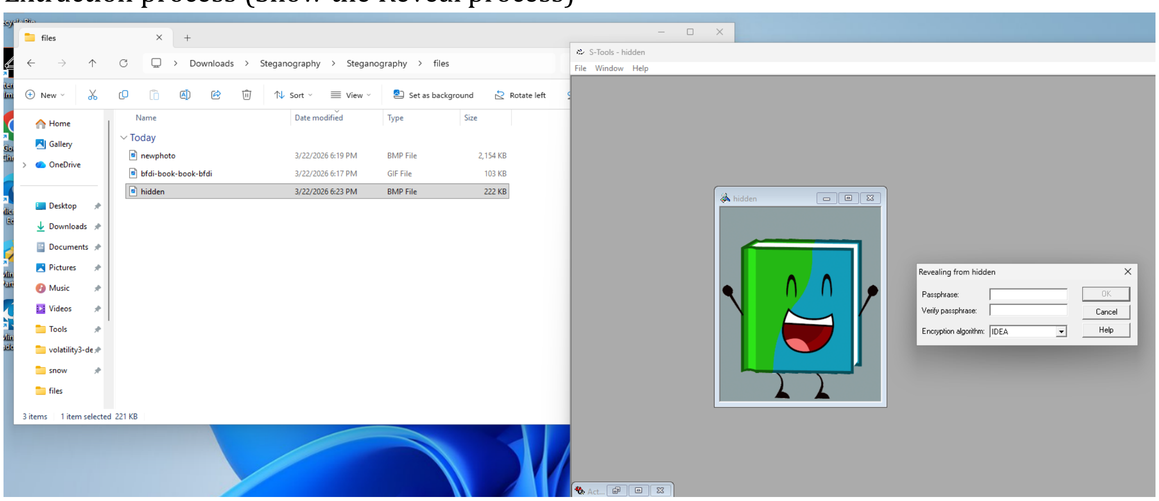
3. Embedding process (show the screen shot where you enter the password for hiding)

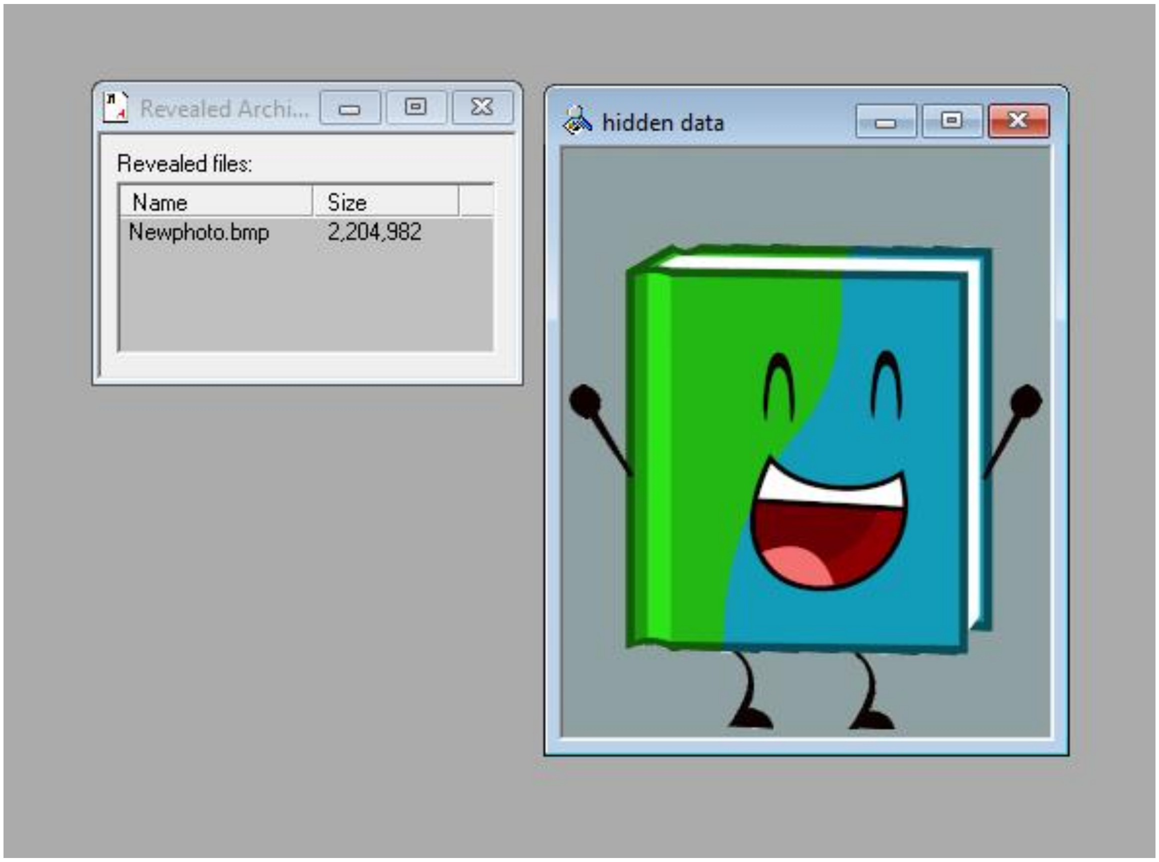


4. Stego image created (show the image created by S-tools)

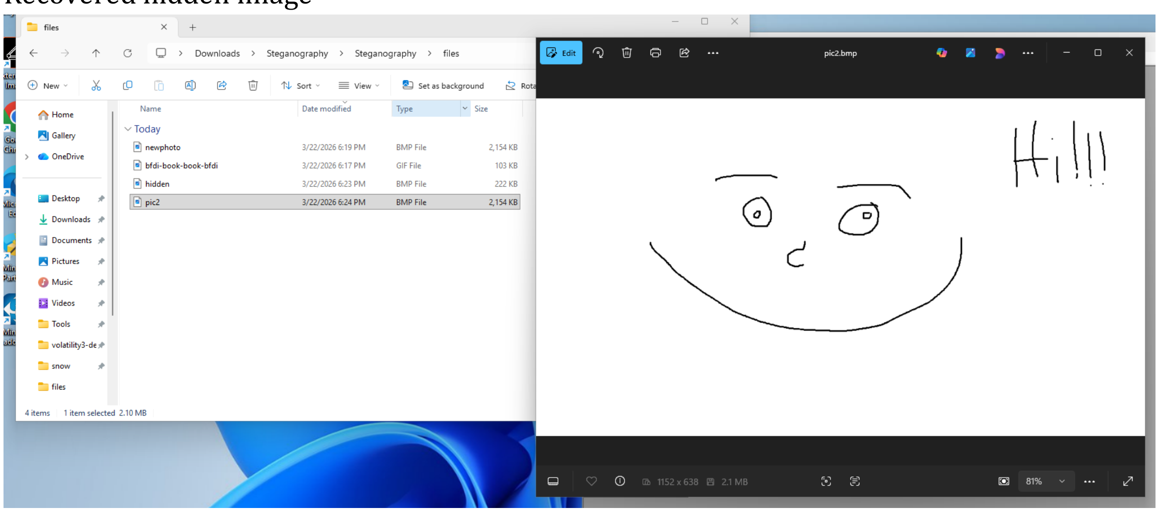


5. Extraction process (Show the Reveal process)

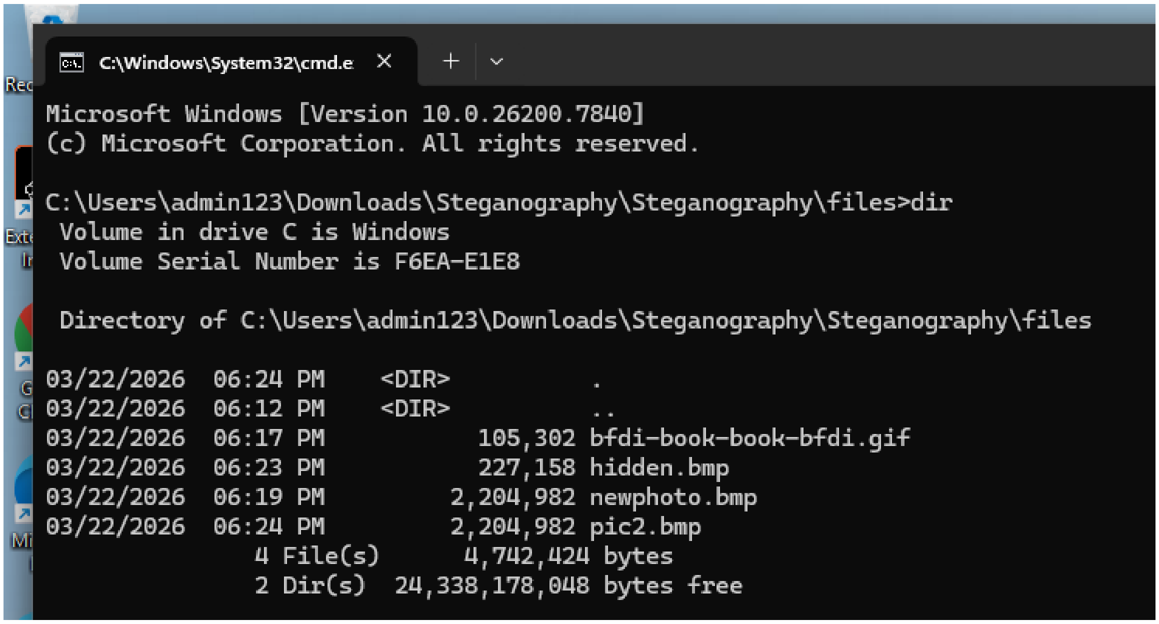




6. Recovered hidden image



7. Output of dir command in command prompt showing the sizes of Stego image file and cover file. Is the size different? Comment on your observation on why and why not the file size has changed.

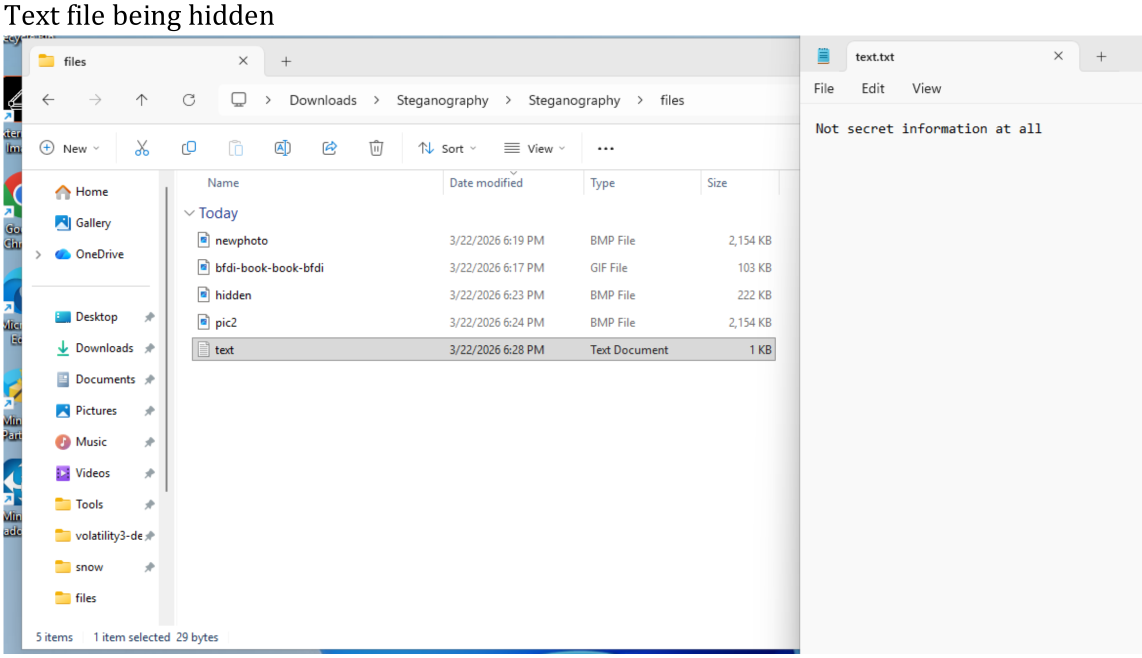


The size of the stego images are much larger than the original cover image due to the hidden data that has been embedded into the image. The data is stored within the least significant bits (LSB) of the image pixels resulting in the noticable file size difference when using the dir command.

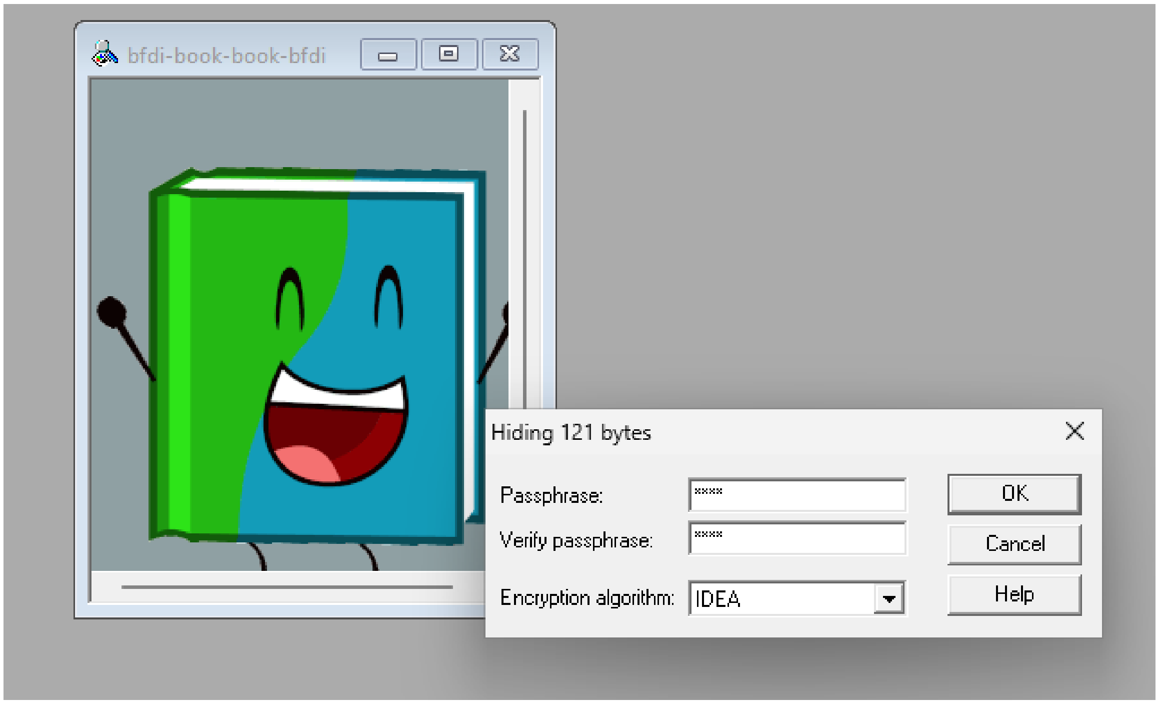
Task 2 – Hide a Text File Inside an Image

Screenshots Required:

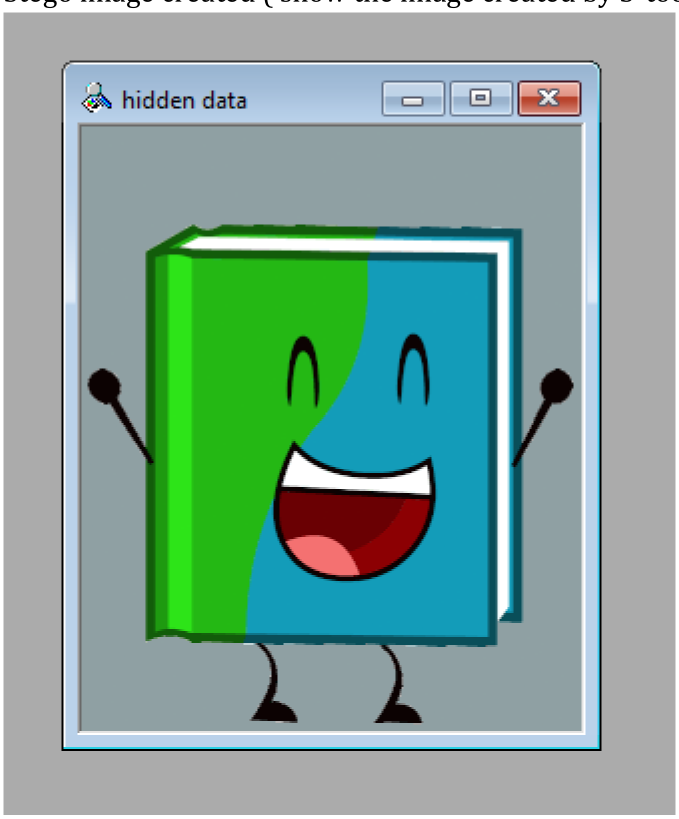
1. Text file being hidden



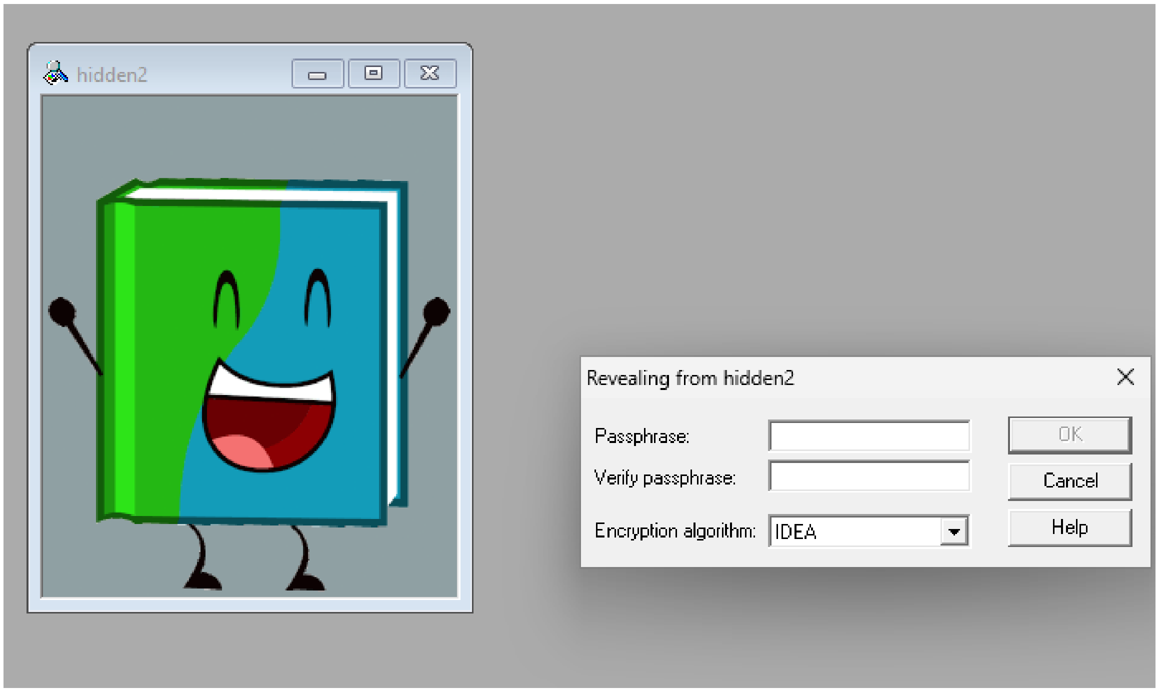
2. Embedding process (show the screen shot where you enter the password for hiding)



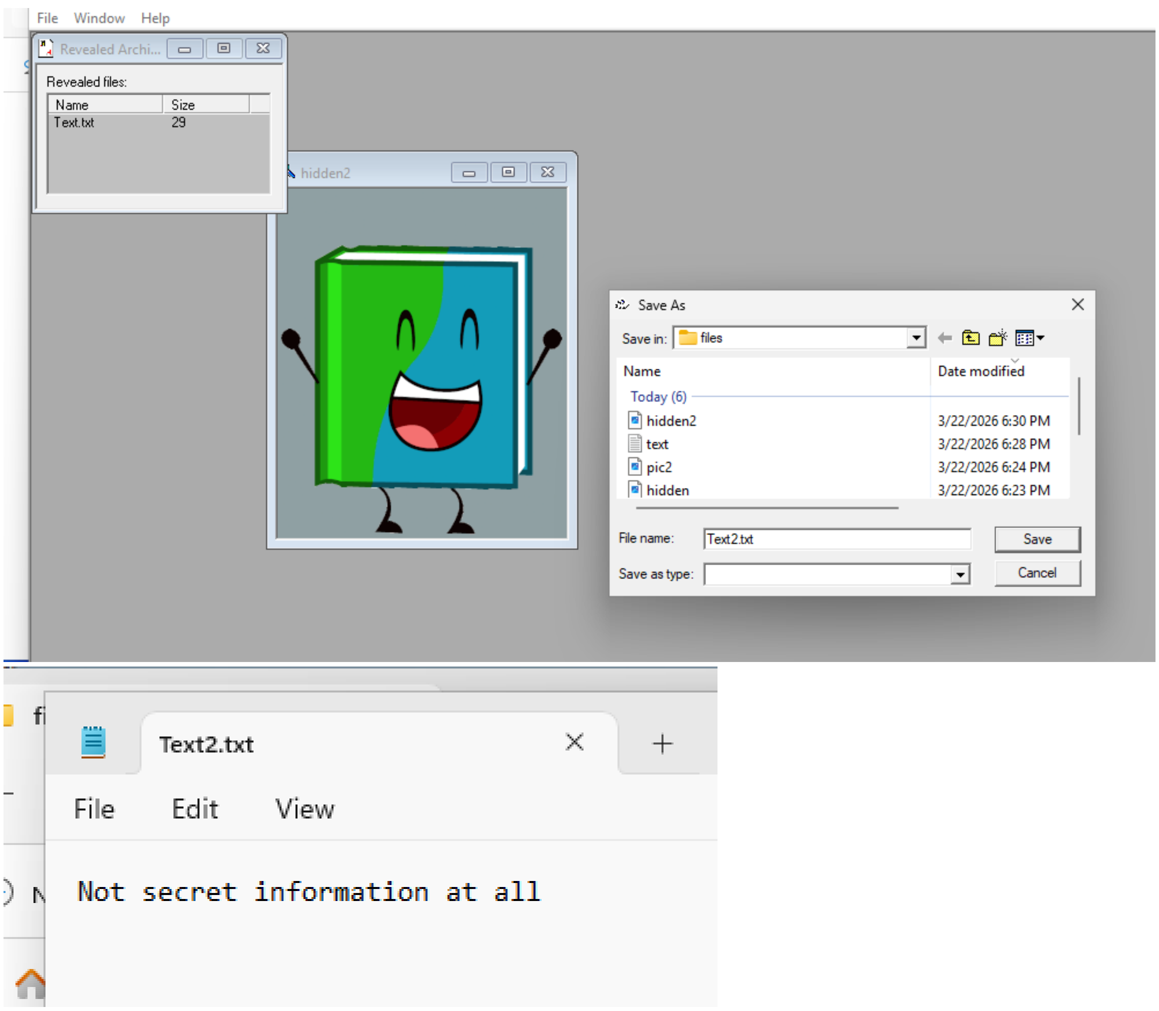
3. Stego image created (show the image created by S-tools)



4. Extraction process (Show the Reveal process)



5. Recovered text file



6. Output of dir command in command prompt showing the sizes of Stego image file and cover file. Is the size different? Comment on your observation on why and why not the file size has changed.

```
 Directory of C:\Users\admin123\Downloads\Steganography\Steganography\files

03/22/2026  06:31 PM    <DIR>          .
03/22/2026  06:12 PM    <DIR>          ..
03/22/2026  06:17 PM           105,302 bfdi-book-book-bfdi.gif
03/22/2026  06:23 PM           227,158 hidden.bmp
03/22/2026  06:30 PM           227,158 hidden2.bmp
03/22/2026  06:19 PM         2,204,982 newphoto.bmp
03/22/2026  06:24 PM         2,204,982 pic2.bmp
03/22/2026  06:28 PM                29 text.txt
03/22/2026  06:31 PM                29 Text2.txt
               7 File(s)      4,969,640 bytes
               2 Dir(s)  24,336,125,952 bytes free
```

The stego file image with the txt embedded is about double the size of the original image file. The method used is the same as when embedding images within eachother.

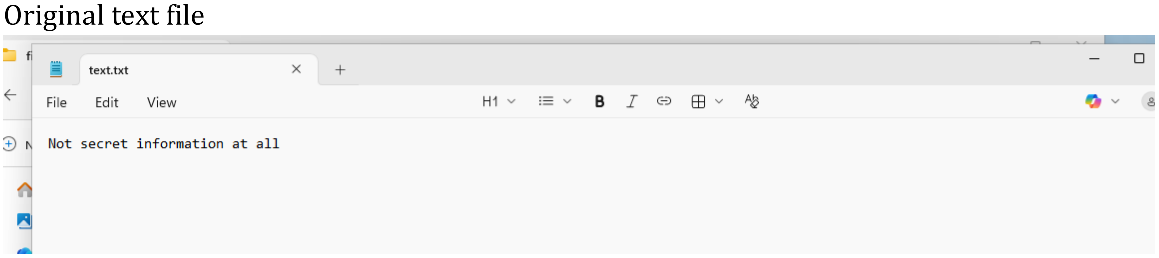
Part 3 – Steganography Using StegoMagic

Follow the Lecture 3 Video posted for this week in Blackboard.

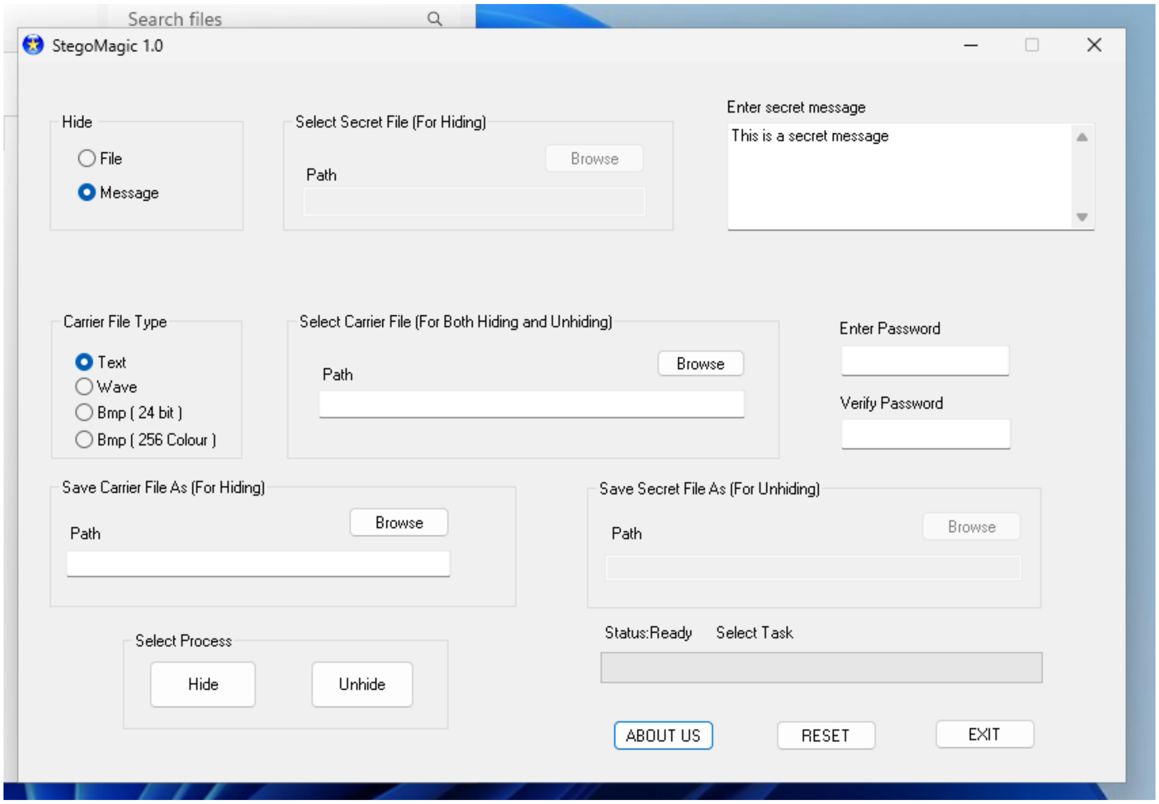
Task: Hide a simple message in a Text file. Create a simple Text file and hide some message in it using the StegoMagic tool.

Screenshots Required:

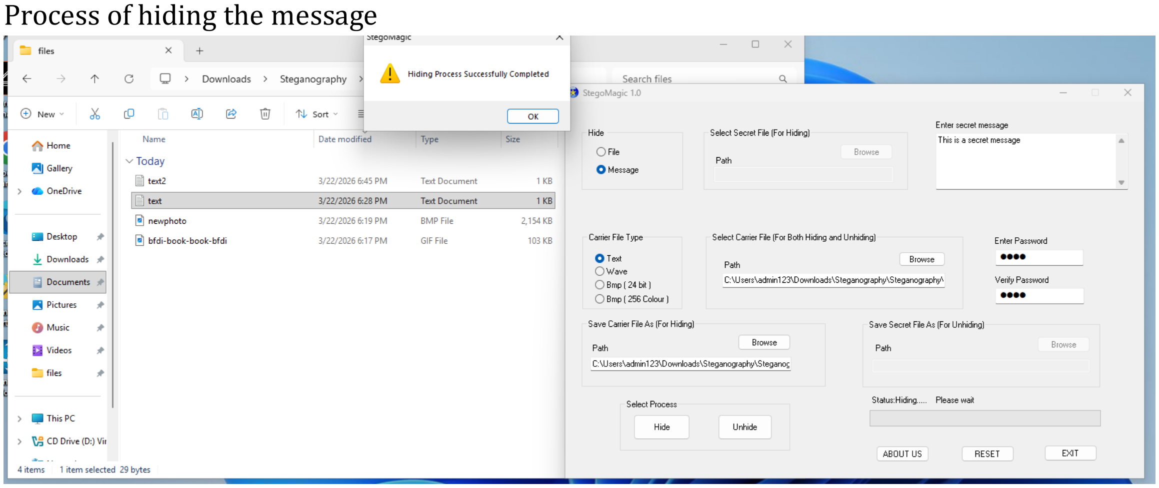
1. Original text file



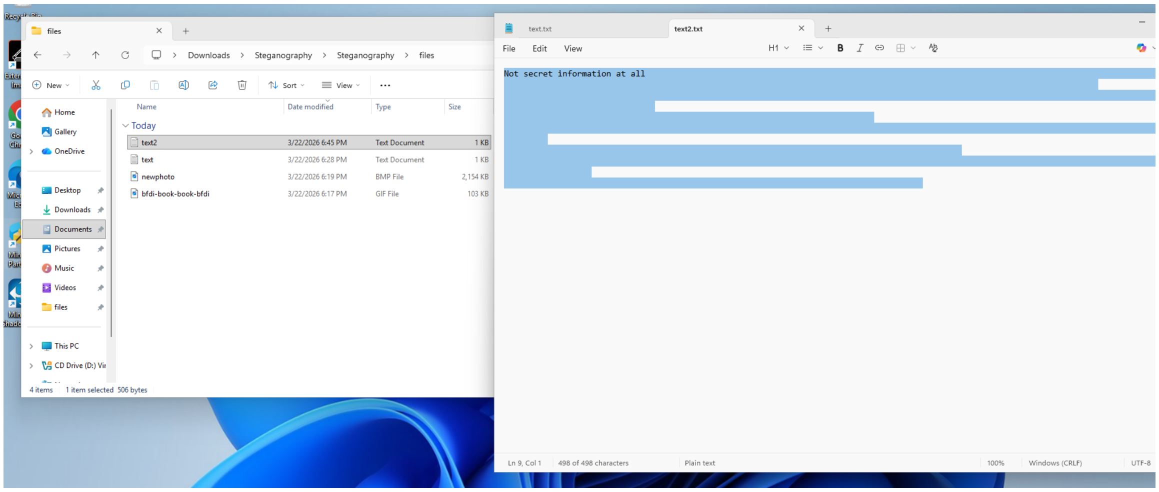
2. Message entered into StegoMagic



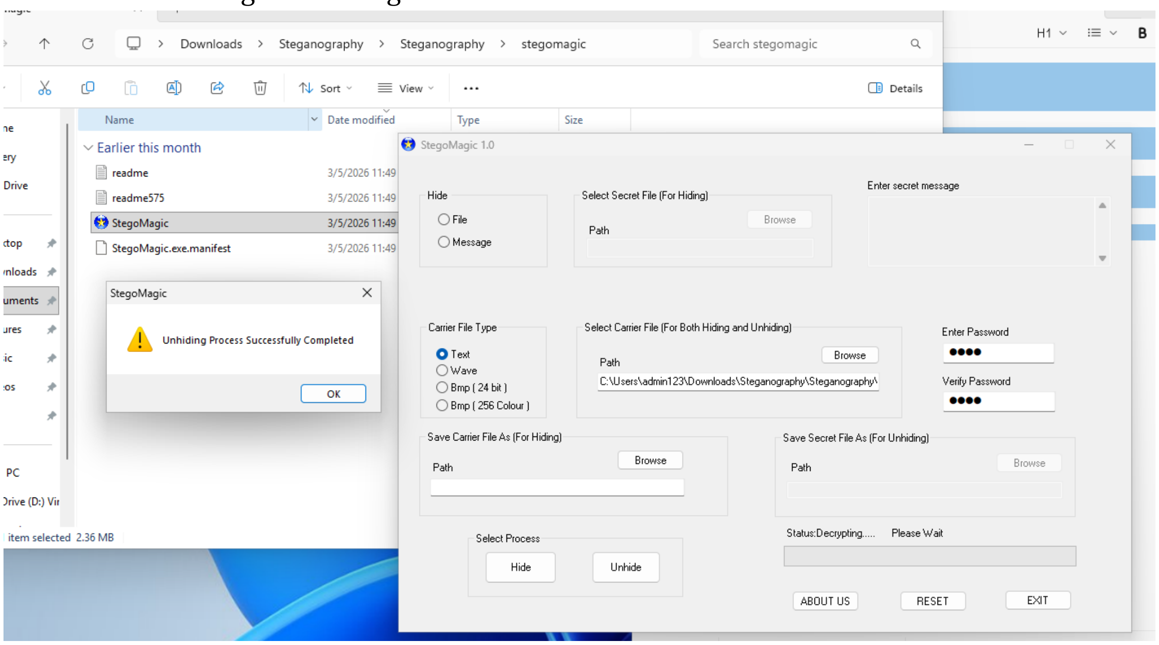
3. Process of hiding the message



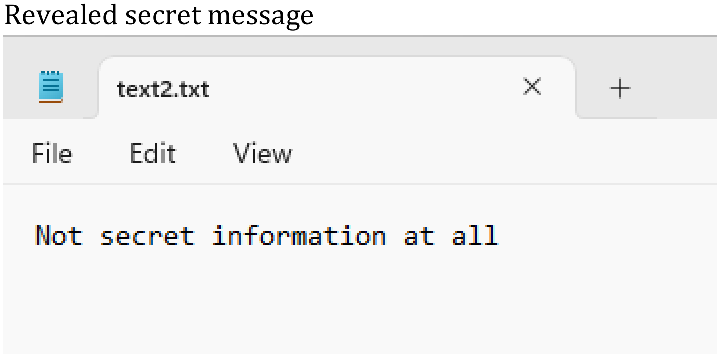
4. Modified file after hiding the message



5. Process of revealing the message



6. Revealed secret message



Lab Submission

Submit a single document containing:

• All required screenshots

- Answers to the questions